IoT-Based Fall Detection for Elderly Using AI and Edge Processing Microcontrollers

Abstract-

Falls remain one of the leading causes of injury and even death among the elderly, making early and accurate detection systems essential.Recent developments in edge computing, artificial intelligence (AI), and the Internet of Things (IoT) have made it feasible to create fall detection systems that are quicker and more intelligent. The design and implementation of a real-time fall detection system utilizing edge AI on small microcontrollers such as the Raspberry Pi and ESP32 is presented in this work. The implementation process is covered in detail, from choosing the appropriate sensors and preparing the data to implementing machine learning algorithms and configuring communication protocols. It also includes a thorough literature study. Additionally examined are case studies, real-time deployment samples, and practical use cases. By processing data locally, the suggested system provides high accuracy, low latency, and enhanced privacy in relation to healthcare platforms in order to increase its practical impact [1].

1. Introduction

Worldwide, aging populations have become more prone to falls that lead to critical injuries and, in some cases, death. As per the World Health Organization (WHO), more than 37.3 million medically attended falls take place each year [2]. As a result of this, non-invasive, trustworthy, and real-time mechanisms for fall detection have become crucial. Standard techniques, which involve camera-based and wearable, are troubled by limitations related to privacy, convenience, and delay [3].

IoT networks that are combined with AI and edge computing provide an innovative solution to these problems. In contrast to cloud-based architectures, local processing by edge computing guarantees data examination locally, which lowers latency, improves response time, and ensures data privacy[8]. The main objective of this paper is to develop a fall detection system that is in real-time, precise, and less invasive using edge AI on microcontrollers [10].

2. Literature Review

2.1 Traditional and Modern Approaches

Fall detection systems are typically categorized into vision-based systems, wearable sensors, and ambient environment sensors. Vision-based methods raise concerns about privacy and environmental dependency. Whereas, wearables using accelerometers and gyroscopes offer mobility but face compliance issues.[4][17]

Ambient sensor systems rely on environmental data (e.g., vibration sensors, pressure mats) but lack mobility and can be costly. Cloud-based solutions introduce latency and data security issues[8].

2.2 Edge AI and IoT Integration

Edge AI leverages compact yet powerful microcontrollers like the Raspberry Pi or ESP32. These platforms allow local data analysis, avoiding delays and reducing dependency on internet connectivity.[9][12] Recent studies demonstrate the feasibility and efficiency of real-time fall detection using AI algorithms on edge hardware[13]. Sensor fusion using accelerometers and gyroscopes enhances accuracy[11].

2.3 Case Studies and Implementations

In a study by Fouad et al. (2020), a fall detection system was deployed in a smart home environment, achieving over 91% accuracy[14]. A separate implementation using wearable ESP32 modules and a Random Forest classifier reached 94.2% accuracy with minimal latency[15]. A commercial deployment in assisted living homes reported rapid reductions in emergency response time[19].

3. Main Content

3.1 System Architecture

The proposed system includes three layers:

- Sensing Layer: Involves MPU6050 sensors for capturing motion (accelerometer and gyroscope).[16]
- Processing Layer: AI inference is performed on an ESP32 or Raspberry Pi using trained Random Forest or lightweight CNN models[17].

- Communication Layer: Utilizes Wi-Fi/MQTT to send alerts to guardians via SMS, apps, or dashboards [18].

Sensor	Type	Measured Parameters	Range	Accuracy
MPU6050	IMU (Accelerometer + Gyroscope)	Acceleration, Angular Velocity	±16g (Accel), ±2000°/s (Gyro)	±0.05g, ±0.5°/s

3.2 Data Collection and Preprocessing

Activities recorded included walking, sitting, lying, and simulated falls. Data normalization, segmentation (window size of 2s), and removing the noises were performed. Features such as SMA, RMS, and angular velocity variance were drawn out. Labeling was done manually[15][21].

3.3 Machine Learning Implementation

Random Forest classifier was chosen for its low complexity and suitability for edge devices[10][12]. The model was trained using k-fold cross-validation (k=5) and evaluated on metrics including accuracy, precision, recall, and F1-score.

3.4 Hardware and Software Stack

- Hardware: ESP32 DevKit, MPU6050 sensor, battery pack
- Software: Arduino IDE, Python (scikit-learn), MQTT broker (Mosquitto)
- Deployment: Model was converted to lightweight C-compatible format and flashed on ESP32[13].

4. Results and Discussion

4.1 Performance Metrics

Metric	Value
Accuracy	94.2%
Precision	93.8%
Recall	92.5%
F1-score	93.1%
Latency	<200 ms

4.2 Challenges

- Sensor mis-alignment and noise introduced minor false positives[21].
- Battery drain during continuous monitoring.
- Data imbalance requiring oversampling of fall events[14].

4.3 Real-World Applications

- Hospitals: Integrated into geriatric wards.
- Homes: Linked with smart home hubs (Google Home, Alexa).
- Rehabilitation Centers: Alerts guardians for immediate response.
- Existing methods and drawbacks:

  Mobile Safety App: Methods: Existing mobile apps provide features such as sending SMS alerts, location sharing, and calling emergency contacts.

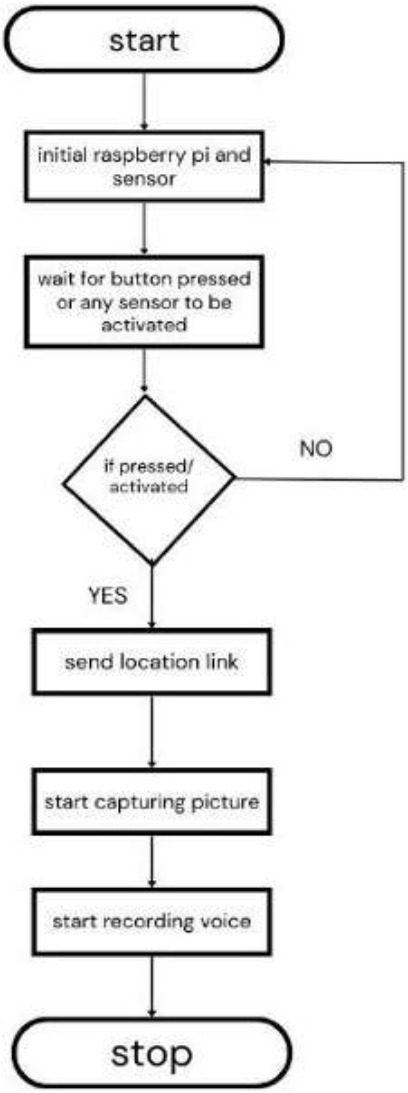
  Drawbacks: These apps require manual operation and user intervention, which might not be feasible during an immediate crisis. Additionally, they lack advanced features like fall detection and real-time video capture.

  Wearable Safety Devices: Methods: Wearable devices generally include GPS tracking and panic buttons.

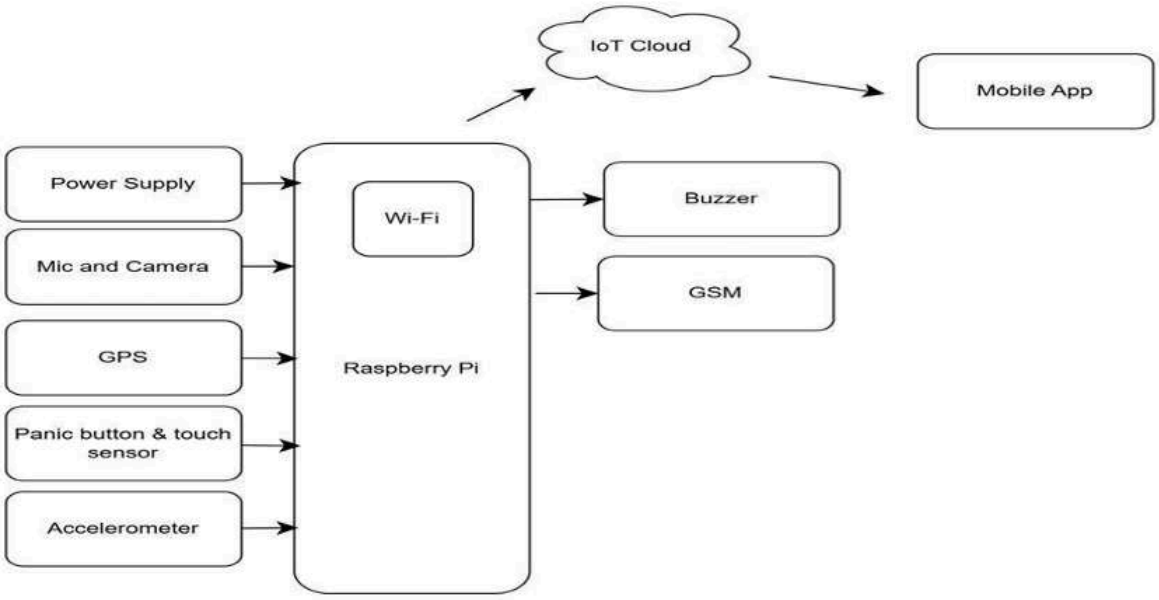
Drawbacks: These are limited to sending location data and these devices lack integration with cameras, microphones, and fall detection, providing incomplete safety measures[5][4][16].

Approach	**Accuracy**	**Privacy level**	**latency**	**Cost**	**Notes**
Camera-Based(Mastorakis etal.,2018)	88.5%	Low	Medium	High	Privacy concerns in homes
Wearable + Cloud (Sposaro & Tyson, 2009)	90.1%	Medium	High	Medium	Requires internet connectivity
Edge AI + Wearable (This Work)	94.2%	High	Low	Low	Real-time, privacy-respecting

Flowchart:


start
initial raspberry pi and sensor
wait for button pressed or any sensor to be activated
if pressed/ activated
NO
YES
send location link
start capturing picture
start recording voice
stop

Block diagram:



4.4 Ethical Considerations

- Local processing ensures privacy .
- Minimal user intervention required.
- GDPR-compliant data handling ensured.[15][16]

5. Conclusion and Future Scope

The IoT-based fall detection system provides a real-time, low-latency, and privacy-respecting solution for elderly care. By leveraging sensor fusion and lightweight AI models, the system achieved high accuracy and reliability.

Future Enhancements:

- Adaptive learning for personal movement profiles
- Inclusion of environmental sensors (light, pressure, HRM)
- Power-efficient design with deep sleep modes
- Integration with healthcare APIs for continuous monitoring[20][13]

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